English Medium

EDU TERIA

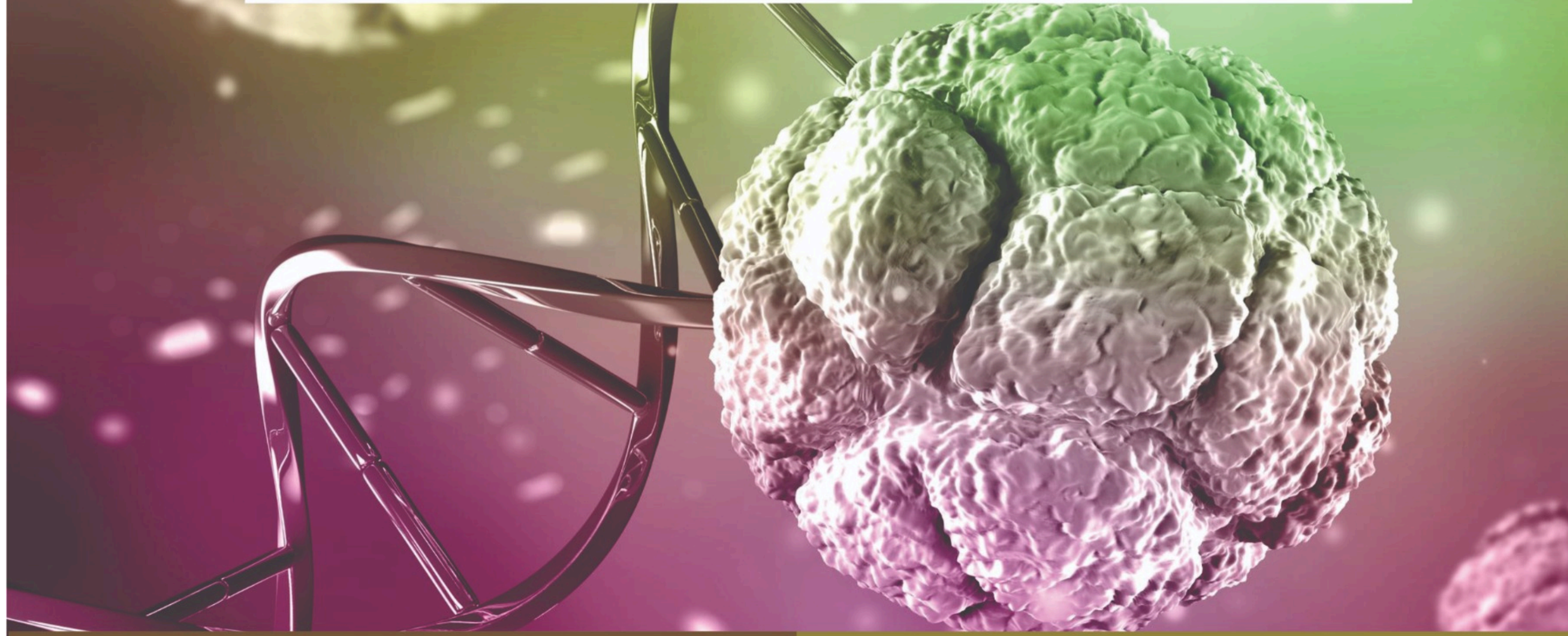
72nd BPSC

Prelims Online Batch

BIOLOGY

LECTURE 8

Plant tissue – Meristem, Permanent tissue, vascular tissue, annual rings.



Plant tissue – Meristem, Permanent tissue, vascular tissue, annual rings

Plant Tissue

Plant tissues are groups of similar cells that work together to perform specific functions in plants. These tissues help in growth, support, transport of materials, and protection of the plant body.

Plant tissues are mainly divided into two major categories:

1. Meristematic tissues (growth tissues)
2. Permanent tissues (differentiated tissues)

Plant Tissue				
Meristematic tissue			Permanent tissue	
Apical meristem	Intercalary meristem	Lateral meristem	Simple permanent tissue	Complex Permanent tissue

1. Meristematic Tissue :

- Meristem or Meristematic tissue is a type of plant tissue composed of actively dividing cells. These cells continuously divide and produce new cells, which help in the growth and development of plants.
- The concept of meristematic tissue was first introduced by Karl Wilhelm von Nägeli.
- Responsible for growth.
- Cells are actively dividing, thin-walled, small, and dense cytoplasm.
- No Intercellular Spaces.

Types of Meristematic Tissue :—

a) Apical Meristem

- Present at root and shoot tips.
- Causes primary growth (increase in height).

Types of Meristematic Tissue :—

b) Intercalary Meristem

- Present at base of leaves or internodes (e.g., grasses).
- Helps in regrowth and elongation.

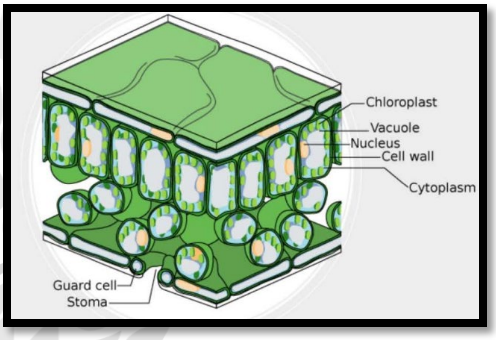
Types of Meristematic Tissue:-

c) Lateral Meristem

- Located on sides of stems and roots (cambia).
- Causes secondary growth (increase in girth).

2. Permanent Tissue:—

- Formed after meristematic cells lose ability to divide.
- Cells may be living or dead.

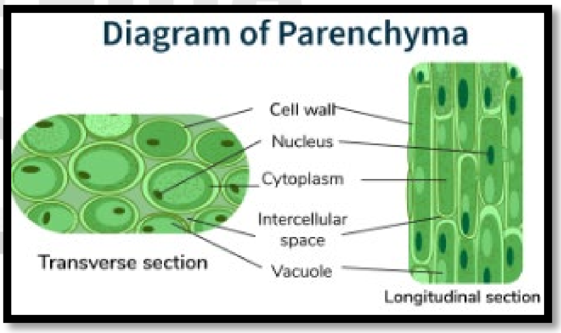


A. Simple Permanent Tissue:—

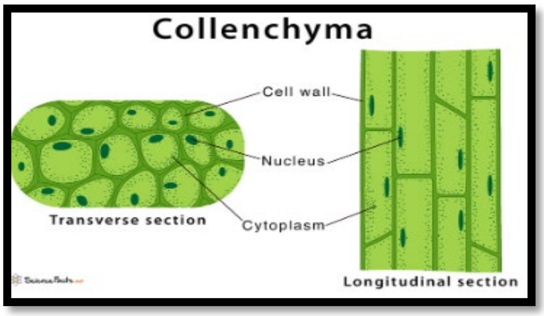
Made of one type of cell.

a) Parenchyma

- Living, thin-walled cells with large vacuoles.
- **Function:** Storage, photosynthesis.
- Aerenchyma (with air spaces) in aquatic plants.



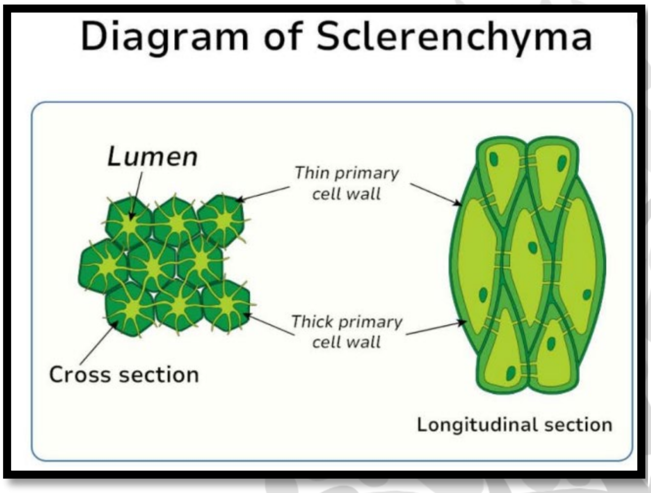
b) Collenchyma

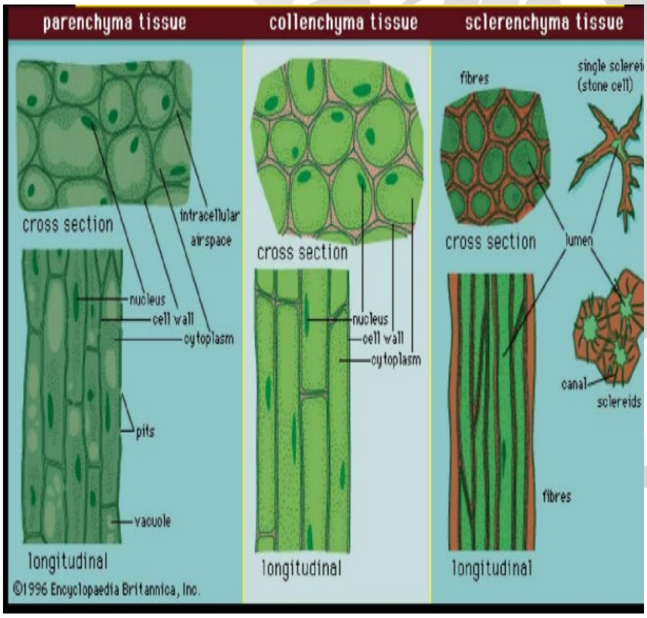


- Living cells with unevenly thickened walls.
- Provides flexibility and young plant..
- Found in leaf stalks (petioles) and below epidermis.

c) **Sclerenchyma**

- Dead tissue with thick, lignified walls.
- Provides strength and rigidity.
- Includes fibers and sclereids (stone cells).





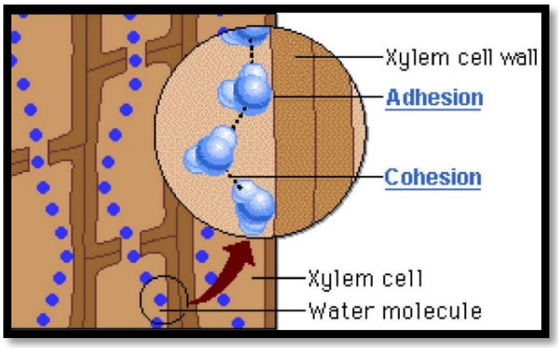
Complex permanent issue:

Made of different types of cells.

a) **Xylem**

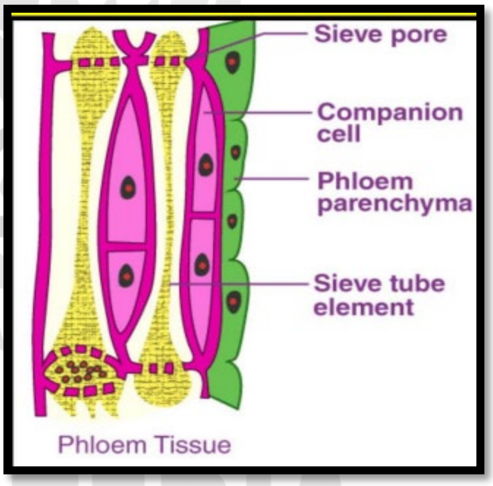
- Conducts water and minerals.
- **Components:** Tracheid's, Vessels, Xylem parenchyma, Xylem fibers.

- Movement is unidirectional...



b) **Phloem**

- Conducts food from leaves to other parts.
- **Components:** Sieve tubes, Companion cells, Phloem parenchyma, Phloem fibres.
- Movement is bidirectional.



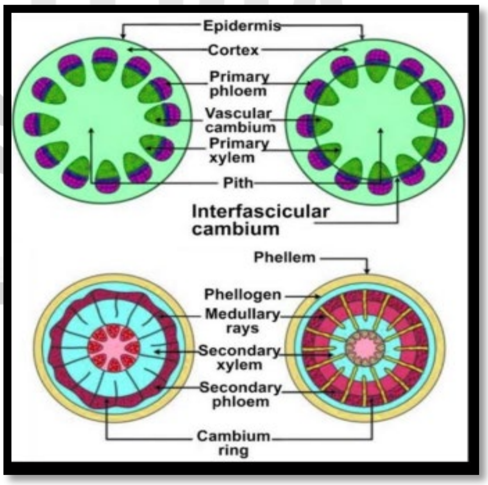
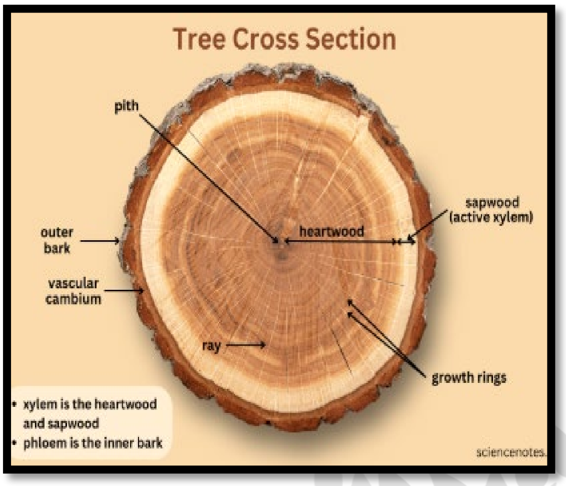


Figure: Diagram of Plant Tissue.

Annual Rings

- Annual rings are circular rings visible in the cross-section of a tree trunk that represent yearly growth of the plant.

- They are formed due to the activity of vascular cambium during secondary growth.



Importance of Annual Rings :

1. Helps determine the age of a tree
2. Provides information about climatic conditions
3. Important in dendrochronology (study of tree rings)